

A Polite, Container-Native Bursting Fabric for AI Workloads on Shared Clusters

Andrew H. Bond
San José State University
San José, CA, USA
andrew.bond@sjsu.edu

ORCID: 0009-0003-2599-6158

Abstract—A researcher with a single workstation and access to a shared, multi-tenant GPU cluster (the National Research Platform’s Nautilus) faces real friction turning that access into elastic capacity: `kubectl`, YAML, container cold starts, and no feedback loop into the code they are already running. We present a *polite, container-native bursting fabric* that makes the cluster’s GPUs appear as elastic capacity *inside the user’s running program*. The fabric, *nats-bursting*, makes containerized cluster pods first-class subscribers on the same NATS bus as the workstation—in two shapes, ephemeral Kubernetes Jobs and persistent queue-group pools—behind a Go control plane whose *politeness controller* probes live cluster state and backs off rather than crowding other tenants, operationalizing shared-cluster citizenship. Its NATS leaf-node topology surfaces real orchestration design issues (JetStream vs. core-NATS delivery, asymmetric subject interest) that we report as an experience. Two supporting components make the fabric practical: *turboquant-pro* compresses the payloads on the wire so bursting stays viable over constrained links, and *batch-probe* right-sizes each pod’s GPU batch so bursts stay useful and policy-aligned. We quantify the fabric on real infrastructure: a containerized burst cold-starts in ~ 6.6 s while a warm pool serves at ~ 67 ms; over a real ~ 2 Mbps link, compression cuts round-trip up to $8.4\times$; and *batch-probe* drives a burst to 94% GPU utilization. We report what is measured and what is not, and release all three projects and the reproducible benchmark harness as an artifact.

Index Terms—cloud bursting, Kubernetes, containers, orchestration, NATS, NRP Nautilus, vector quantization, KV-cache compression, GPU utilization, thermal control

I. INTRODUCTION

Shared, multi-tenant Kubernetes clusters such as NRP Nautilus [1] exist precisely so that researchers without their own datacenter can burst. Yet “use the cluster from my workstation” remains awkward: one writes Kubernetes manifests, submits Jobs, polls for completion, and shuttles data in and out—none of it inside the program where the research runs. The friction is high enough that elastic capacity often goes unused.

Consider a concrete case. A researcher iterating on an embedding or inference pipeline at their workstation wants, for one hour, to fan ten thousand documents across a dozen cluster GPUs, then return to interactive work. Today that means authoring a Job manifest, building and pushing a container image, submitting, polling, and copying results back—a context switch out of the program and into cluster operations. The capability exists; the ergonomics do not. And on a *shared* cluster a second obligation rides along: a convenience layer

must not crowd out the other tenants, or it will (rightly) get its owner throttled or banned.

This paper describes a polite, container-native *bursting fabric* that closes the gap while honoring that obligation, and measures it honestly on the real cluster. Its core is the fabric itself; two supporting components make remote bursting practical:

- **The bursting fabric** (*nats-bursting*, §III)—the primary contribution: a Go control plane plus a Python client that make containerized cluster pods *first-class subscribers* on the researcher’s own NATS bus, in two shapes (ephemeral Jobs, persistent pools), behind a *politeness controller* that probes cluster state and backs off to yield to other tenants. Its NATS leaf-node topology surfaces concrete orchestration design tradeoffs (JetStream vs. core-NATS delivery, asymmetric subject interest) we report as an experience.
- **Transport optimization** (*turboquant-pro*, §IV)—compressing bus payloads so remote bursting stays viable over constrained links, reported at the level a practitioner needs: ratios, fidelity, real-link speedup, and a rule for *when* compression helps.
- **Resource management** (*batch-probe*, §V)—right-sizing each pod’s GPU batch (and thermally governing the local driver) so bursts remain useful and within a shared cluster’s utilization policy.

We then quantify the fabric on real infrastructure (§VI), state threats to validity (§VII), and reflect on the engineering (§VIII). This is a practice/experience paper: a working, open system plus measurements. Its value is the orchestration design—particularly the politeness controller that makes “effective and policy-safe” mechanical rather than aspirational—and the honest characterization of when each supporting layer helps.

II. BACKGROUND

NRP Nautilus [1] is a shared, opportunistic Kubernetes cluster. Its usage policy shapes the design: small/idle pods are tolerated ($\text{CPU} \leq 1$, $\text{mem} \leq 2$ GiB are an “ignored” range), but groups of pods must sustain utilization (5+ pods must each stay under 40% GPU), Jobs must run to completion and exit cleanly (no *sleep-camping*), and requests should match

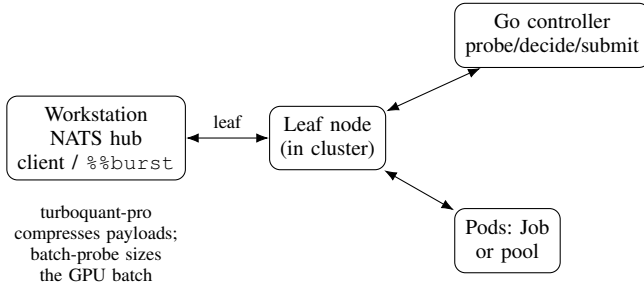


Fig. 1. Two shapes, one bus. Ephemeral bursts go workstation→Go controller→Kubernetes Job; persistent pools are queue-group worker Deployments the client dispatches to directly. Both reach the workstation over an outbound leaf link.

usage. “Effective and policy-safe” is therefore a container-orchestration design target, not a slogan.

NATS [2] provides subject-based publish/subscribe and request-reply; *queue groups* that round-robin a subject across a set of subscribers; *JetStream*, a persistence layer with work-queue streams and durable pull consumers (at-least-once, redelivery); and *leaf nodes* that bridge a remote NATS server into a hub. Leaf-node federation is central to our design and to two of its sharper engineering decisions (§III): JetStream is not federated across a leaf link, and subject *interest* can propagate asymmetrically hub→spoke.

Cloud bursting spills work from a local resource to a remote one. We use a message bus rather than a job queue or RPC because it makes the containerized cluster a *peer* on the fabric the application already speaks, so a burst becomes a publish inside the event loop.

III. NATS-BURSTING: THE BURSTING FABRIC

nats-bursting has two halves joined by one bus (Fig. 1): a Go control plane that lives next to the cluster and turns NATS messages into Kubernetes objects, and a Python client used from notebooks and scripts. Two workload *shapes* share the bus but use different control paths.

A. The control plane and its politeness controller

The Go controller is event-driven. Its `natsbridge` subscribes to `burst.submit`, and on each request the submitter renders a `JobDescriptor` into a real `batchv1.Job` via `client-go`—CPU/memory/nvidia.com/gpu/ephemeral-storage requests and limits, a managed-by label, `RestartPolicyNever`, and foreground-propagation cancellation—then publishes a status event. The wire protocol is three subject families: `burst.submit` (requests), `burst.status.<job>` (a submitted/error state machine), and `burst.result.<job>` (reserved for results). The client issues a request-reply on `burst.submit` and falls back to subscribing on the status subject—robust to whichever pattern the controller answers with.

The part the rest of this paper’s “policy-safe” claim rests on is the *politeness controller*. A *prober* reads live cluster

Algorithm 1: Polite admission decision

Input: cluster state s (from the prober), thresholds p
Output: Submit / Backoff / Abort
if attempts $\geq p.maxAttempts$ **then return** Abort;
if $s.myRunning \geq p.maxConcurrent$ **then return** Backoff;
if $s.myPending \geq p.maxPending$ **then return** Backoff;
if $s.othersPending > p.queueDepth$ **then return** Backoff;
if $s.alloc/s.total > p.util$ **then return** Backoff;
return Submit;

TABLE I
POLITENESS CONTROLLER DEFAULTS.

parameter	default
max concurrent jobs (self-limit)	50
max pending jobs (self-limit)	20
others-pending threshold (courtesy)	200
utilization threshold (load)	0.90
initial / max backoff	30 s / 30 min
backoff multiplier / max attempts	2.0 / 20

state through `client-go`—total/allocated/idle nodes, the caller’s own running and pending Jobs, *and other tenants’ pending pods*—and degrades gracefully when an individual API call is denied by RBAC. A *decider* then evaluates that state against thresholds in a fixed priority order: (1) *self-limiting*—do not exceed your own running/pending caps (defaults 50/20); (2) *courtesy*—yield when more than a threshold of *other users’ pods* are queued (default 200); (3) *utilization*—back off when cluster allocation exceeds a fraction (default 0.90). The outcome is Submit/Backoff/Abort (Algorithm 1, with thresholds in Table I). Backoff is capped exponential with jitter applied *after* the cap,

$$w_k = \min(B_0 m^k, B_{\max}) \cdot \left(1 + \frac{1}{2} \left(u - \frac{1}{2}\right)\right), \quad u \sim \mathcal{U}[0, 1), \quad (1)$$

i.e. $\pm 25\%$ jitter to decorrelate wake-ups across many controllers (anti-thundering-herd), aborting after K attempts. This is what makes “yield to other tenants” a mechanism rather than a promise: the system measures the shared cluster and defers to it.

B. The Python client and the %%burst magic

The client is a synchronous facade over async NATS, owning one background event loop and a Transport seam for deterministic testing. `JobDescriptor/Resources` mirror the Go structs field-for-field. A `%%burst` IPython cell magic makes a burst an editor action: by default (`-when-busy`) it probes the local GPUs through `nvidia-smi` and only offloads when every local GPU is saturated, otherwise running the cell in place; `-always/-never` force the choice, and `-gpu/-cpu/-image` set resources. The cell body is shipped

as an environment variable and executed in the pod, so no image rebuild is needed for ad-hoc work.

C. Persistent pools, queue groups, and leaf federation

The second shape is a persistent pool: a Deployment of always-on workers that subscribe as a NATS queue group and stay warm (no per-task cold start, cached models). A `TaskDispatcher` submits many tasks and collects results on `results.>`. Two delivery models coexist, and the choice is a deliberate accommodation of leaf-node topology. With `durable=True`, tasks go through a JetStream work-queue stream consumed by a durable pull consumer shared across replicas (at-least-once, explicit ack/nak, bounded redelivery)—correct when dispatcher and workers share a JetStream domain. With `durable=False`, tasks use core NATS queue groups, which cross leaf-node boundaries that JetStream does not. A symmetric subtlety appears on the return path: where subject interest propagates only hub→spoke, results published spoke-side never reach the hub, so the worker can optionally POST results to a webhook that re-publishes them on the hub. The pool’s defaults are NRP-quota-aware (`cpu=1/mem=2Gi` to stay in the “ignored” swarm range), and the worker blocks inside `fetch()` rather than sleeping—satisfying the “no sleeping Jobs” rule—with an idle-exit watchdog that lets the same image run as either a Deployment (run forever) or a self-terminating Job (exit after idle). These are small decisions, but each is the difference between a demo and something that survives contact with a real multi-tenant cluster.

IV. TRANSPORT: MAKING BURSTING VIABLE OVER CONSTRAINED LINKS

When work bursts to a remote cluster, the payloads—embeddings, KV-cache pages, feature tensors—ride the bus, and on a constrained link they, not the cluster, become the bottleneck (§VI). `turboquant-pro` compresses them so remote bursting stays viable; we report only what a deployer needs, not the codec’s internals.

A distribution-free codec. Each vector is L2-normalized, rotated by a shared, seed-deterministic random rotation, and each rotated coordinate is quantized against a fixed Lloyd–Max codebook and bit-packed. The rotation is the trick: by concentration of measure it makes the coordinates approximately i.i.d. Gaussian *regardless of the input distribution* [3], so one fixed codebook is near-optimal for any data with no training or calibration. The NATS codec frames each embedding as an 8-byte header (version, bits, dim, norm) plus $\lceil \text{dim} \cdot \text{bits}/8 \rceil$ packed bytes; the same construction serves KV-cache *values*, while *keys*—whose error the softmax amplifies—instead use a per-channel scheme (per-vector normalization is near-lossless for values but catastrophic for keys), and values support attention computed directly on the compressed codes [4], [5], though that kernel-level work is out of scope here.

What a deployer needs. Compression is 8–16× at 4/3/2 bits with reconstruction cosine 0.94–0.995, and is *distribution-agnostic*—real transformer embeddings reproduce the random-vector numbers to within 0.001 (§VI). The deployment rule is

Algorithm 2: OOM-safe batch right-sizing (infer mode)

```

Input: model  $M$ , input_fn, range  $[lo, hi]$ , headroom  $h$ 
 $best \leftarrow 0$ ;
while  $lo \leq hi$  do
   $b \leftarrow \lfloor (lo + hi)/2 \rfloor$ ; cleanup();
   $ok \leftarrow \text{forward}(M, \text{input\_fn}(b))$  completes
    without OOM;
  free trial tensors; cleanup();
  // unconditional: release fragments
  if  $ok$  then  $best \leftarrow b$ ;  $lo \leftarrow b+1$ ;
  else  $hi \leftarrow b-1$ ;
return  $\max(1, \lfloor best(1-h) \rfloor)$ ;

```

simple and we evidence it: *compress when the link or bus is the bottleneck*. On a fast LAN it is a wash; on a real ~2 Mbps link it cuts NATS round-trip up to 8.4× for large payloads (§VI). Encode/decode are cheap relative to a constrained link but not free, so the rule—not a blanket “always compress”—is what we recommend.

V. RESOURCE MANAGEMENT: KEEPING PODS USEFUL AND POLICY-ALIGNED

A burst is only worth its slot if each pod actually fills its GPU—a shared cluster rewards high utilization and penalizes hoarding—and only sustainable if the node driving the burst does not overheat. `batch-probe` provides both, as a support mechanism for the fabric.

A. OOM-safe batch right-sizing

`probe_batch_size` binary-searches the largest batch that fits, building inputs through a caller-supplied `input_fn(b)` and running a trial at each candidate. In *infer* mode the trial is a forward pass under `no_grad`; in *train* mode it instantiates an AdamW optimizer and runs a full forward/backward/step, so the search reflects *true* peak training memory including optimizer moments—a detail most batch finders miss and under-estimate by roughly the optimizer-state factor. OOM is caught and distinguished from genuine errors by message inspection; an unconditional `finally` block deletes trial tensors and flushes the allocator before the next iteration, which is what keeps the search stable across repeated OOMs (Algorithm 2). Headroom is a multiplicative shrink of the largest success (default 0.8×).

A generic `probe(work_fn)` extends the same recovery to any framework (Torch/CuPy/JAX) by abstracting the memory-pool flush.

B. Thermal-aware throttling

On the local orchestration side, `ThermalController` keeps the workstation driving a burst within a thermal budget. Rather than a fixed temperature threshold, it estimates thermal

state with a two-state Kalman filter [6] (temperature and its rate) and acts on the *predicted* trajectory: when it forecasts an overshoot within a short horizon it throttles worker-thread count *before* the limit is reached, using the filter’s rate estimate as a clean derivative. The same temperature oracle backs a one-shot thread-capacity search and an admission-gating job manager. We keep this brief deliberately—it is a support mechanism here, not the contribution—and note that thermal sensing is currently Linux-only.

VI. EVALUATION

We evaluate the fabric in three tiers by the infrastructure each needs, plus the GPU-sizing demonstration. Every number below is *our own* measurement, reproducible from the released harness (§X); we do not re-run the supporting projects’ internal benchmarks, and report only what bears on the bursting use case.

Experimental setup. The workstation is a remote client. The on-premises node (“Atlas”) has two NVIDIA GV100 32 GB GPUs; it runs the NATS hub and the Tier-1/2 measurements. The shared cluster is NRP Nautilus (namespace `ssu-atlas-ai`), reached from Atlas via `kubectl`, with a leaf-node pod bridging the cluster to the hub. The Tier-2 “real hop” traverses a wide-area overlay link (~ 90 ms RTT, ~ 2 Mbps effective). All cluster runs use ignored-range pods (CPU = 1, mem = 2 GiB) and at most four concurrent pods, to stay within policy.

Measurement protocol. Latencies are p50/p99 over 40–500 requests after warm-up; throughput is the wall-clock rate over the batch. Tier-2 round-trips are timed on the requesting host with a single clock; for the real hop the responder runs on Atlas and the requester on the remote workstation with *no* co-located subscriber, so every request crosses the wire. Tier-3 cold-start is timed from Job submission to the Job’s completion condition, with the schedule→container-start split read from the pod’s own Kubernetes timestamps. Codec encode/decode is timed in isolation (Tier 1) and excluded from the transport loop.

A. Compression (Tier 1)

On dim-1024 vectors the codec gives ratios of 7.9/10.4/15.5 $\times$ at 4/3/2-bit with reconstruction cosine 0.995/0.978/0.94 (Fig. 2). Re-running on real `bge-small-en-v1.5` embeddings (20newsgroups, $n=3000$, native dim 384) gives *identical* ratio and cosine within 0.001 at matched dimension—confirming the distribution-agnostic property empirically. The analytic transfer model follows: a 1 MB batch moves in ~ 84 ms at 100 Mbps but ~ 8 ms compressed (a ~ 63 ms net saving after encode) and is a wash at 1 Gbps—the hypothesis Tier 2 tests. Cosine is a fidelity proxy, not a downstream-task guarantee.

B. Transport (Tier 2)

On a loopback bus (Table II) the $\sim 10\times$ smaller payload already gives up to $\sim 6\times$ lower round-trip for large batches—payload size dominates even with no network. The result loopback cannot show is the *real hop* (Table III, Fig. 3): over a real

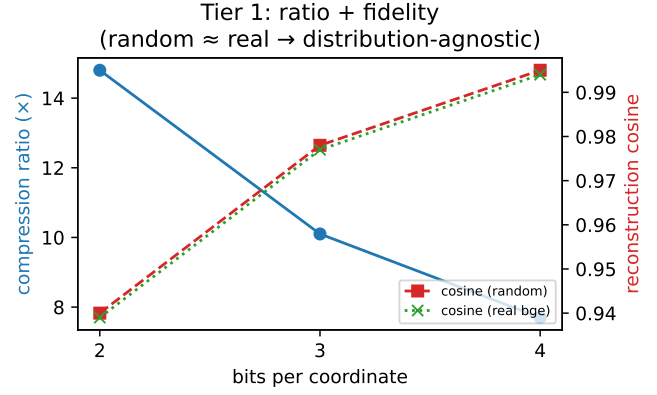


Fig. 2. Tier 1: compression ratio and fidelity vs. bit-width; real embeddings track random within 0.001 cosine.

TABLE II
TIER 2 LOOPBACK (3-BIT; P50/P99 MS, MSG/S).

batch	raw	compressed
1	0.52 / 1.01 / 1768	0.38 / 0.90 / 2261
32	1.69 / 3.12 / 561	0.40 / 1.37 / 2127
256	6.99 / 11.34 / 137	1.20 / 2.61 / 783

~ 2 Mbps link the win scales with payload, from $1.2\times$ (4 KB, latency-bound) to $8.4\times$ (256 KB, link-bound), approaching the raw ratio as the link dominates. This demonstrates the crossover on a real link. Encode cost is excluded from the loop (it is in Tier 1), so the speedups are not free.

C. Cluster: cold-start, warm-pool, scaling (Tier 3)

Ephemeral CPU Jobs (ignored range, one at a time, auto-cleaned) cold-start in median ~ 6.6 s (5.1–7.6 s, $n=5$), of which the K8s schedule→container-start is ~ 1 –3 s. A persistent pool of N queue-group workers driven from the hub (Table IV, Fig. 4) serves at ~ 67 ms per task—removing the $\sim 100\times$ cold-start penalty. Throughput plateaus at ~ 930 /s by two workers, and honestly so: for trivial I/O-bound tasks the ceiling is concurrency $\div$ RTT (Little’s law, $64/0.067 \approx 955$ /s), not worker count—one to two pods saturate the WAN link. Worker-count scaling is expected to matter for *compute-bound* tasks. Finally, batch-probe sized a BERT-base-scale encoder to batch 204 on a GV100 and the burst ran at **94.4% mean / 100% peak GPU utilization** at a safe 61°C—far above the $>40\%$ policy expectation. (The batch is bounded by a safety-capped search, not device memory; the utilization is the result.)

VII. THREATS TO VALIDITY

We state these so the numbers are read correctly. **Synthetic data (Tier 1):** real bge embeddings reproduce the random numbers within 0.001; cosine remains a fidelity proxy. **Loopback (Tier 2):** addressed by the real ~ 2 Mbps hop ($1.2 \rightarrow 8.4\times$); it is one overlay (likely relayed) path—a characterized datacenter link would pin absolute bandwidth. **Cold-start realism (Tier 3):** warm node + cached image +

TABLE III
TIER 2 OVER A REAL ~ 2 MBPS HOP (3-BIT, P50).

batch	raw	compressed	speedup
1	4 KB / 85 ms	0.4 KB / 71 ms	1.2 $\times$
16	64 KB / 383 ms	6 KB / 105 ms	3.6 $\times$
64	256 KB / 1119 ms	24 KB / 134 ms	8.4 $\times$

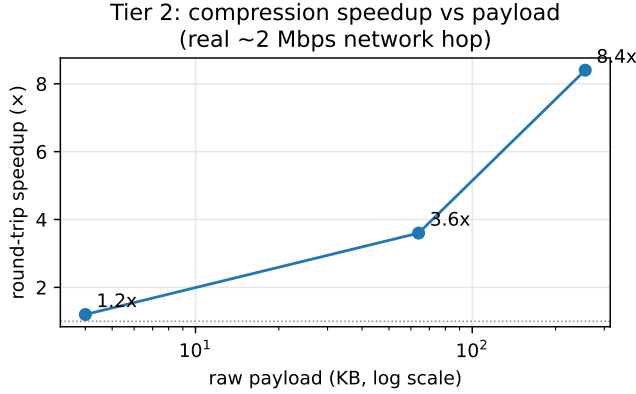


Fig. 3. Tier 2: compression speedup grows with payload on a real bandwidth-limited link.

kubectl wait inflate the total; the K8s breakdown isolates the cluster-side cost; warm-pool and a real NATS-join are measured, while a cold image pull and a GPU-image run remain. **Scaling:** the plateau is explained by Little’s law, not presented as pod-count scaling; a compute-bound load is the named next run. **Portability:** batch-probe’s thermal sensing is Linux-only and its feedforward is one-sided. **Cluster citizenship:** the harness is policy-safe by construction (ignored-range pods, ≤ 4 concurrent, exit-0 Jobs, TTL + delete cleanup, an explicit policy gate, and—above the harness—the controller’s politeness backoff).

VIII. DISCUSSION

“Effective” use is batch-probe sizing pods to high utilization (94% measured); “policy-safe” is the politeness controller plus the gate, short runs, and cleanup. The deployment rule for compression is now evidenced: compress when the link or bus is the bottleneck—a wash on a fast LAN, a large and growing win on a real WAN-class link.

Lessons learned. Three points generalize. First, on a multi-tenant cluster the *usage policy is a first-class design constraint*: it shaped the politeness controller, the ignored-range defaults, and the no-sleep worker as much as any performance goal, and a benchmark that strays outside policy is itself a bug. Second, *leaf-node federation leaks into the application*: the durable-vs-core dispatch choice and the result webhook both exist only because JetStream and subject interest do not cross a leaf link symmetrically—a from-scratch design that ignored this would deadlock on the return path. Third, *honest negatives beat flattering curves*: reporting the Little’s-law plateau and labeling the cold-start as warm-node/cached-image makes the

TABLE IV
TIER 3 WARM-POOL + SCALING (1 KB TASKS, CONCURRENCY 64).

workers	throughput	p50	p99
1	782/s	71 ms	113 ms
2	924/s	68 ms	80 ms
4	936/s	67 ms	78 ms

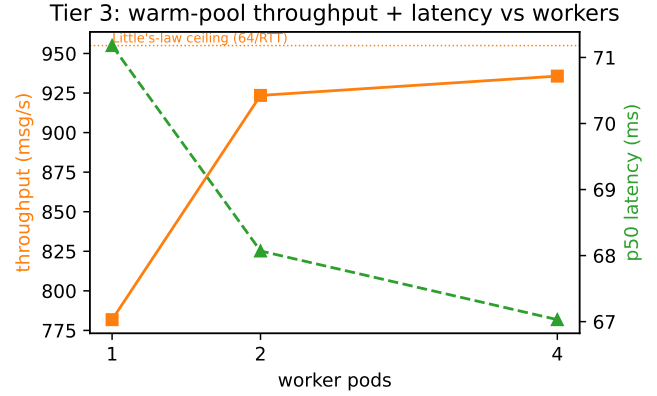


Fig. 4. Tier 3: warm-pool throughput and latency vs. workers; throughput is bounded by concurrency $\div$ RTT, not pod count, for trivial tasks.

result more useful to an operator than a scaling plot that does not hold.

When to use which shape. An ephemeral Job costs a ~ 6.6 s cold start but holds no resources between bursts—right for occasional or embarrassingly parallel work. A persistent pool pays that once and serves at ~ 67 ms but occupies its slots—right for interactive or sustained workloads, provided it is sized within policy and torn down when idle. The bus makes switching a configuration choice, not a rewrite.

IX. RELATED WORK

Bursting and task frameworks. Spilling local work to remote capacity is long-studied: HTCondor flocking [7] federates pools; Globus Compute/funcX [8] offers a federated function-serving fabric; Ray [9], Dask [10], and Parsl [11] build task graphs over remote workers. All treat the remote resource as an external system one submits to and polls. Message-oriented task systems (Celery, RabbitMQ workers) put a broker between producer and consumer, and Kubernetes-native batch layers (Kueue, Volcano) schedule Jobs beneath the application. Two things set our approach apart: *orientation*—containerized pods become first-class subscribers on the researcher’s own bus, so a burst is a publish inside the event loop rather than a hand-off—and an explicit *politeness* control loop that probes a shared, multi-tenant cluster and backs off to yield to other tenants, which the submit-and-poll model leaves to the user.

Quantization and attention. turboquant-pro builds on the random-rotation, near-optimal-distortion line of TurboQuant [3] and the broader embedding/KV-cache quantization

TABLE V

SHIPPED IMPLEMENTATION PER PROJECT (IMPORTABLE PACKAGE OR BINARY; EXCLUDES TESTS, BENCHMARKS, AND EXPERIMENT SCRIPTS).

project / component	language	LOC
nats-bursting controller	Go	752
nats-bursting client + pools	Python	1,398
turboquant-pro	Python (+CUDA via CuPy)	11,379
batch-probe	Python	1,082
total		14,611

literature; its compute-on-codes fused decode applies online-softmax and IO-aware attention techniques [4], [5] in the *code* domain so that quantized values are never reconstructed (per-channel keys are reconstructed). Our paper does not propose a new codec; it contributes a measured account of *when* on-wire compression pays for the bursting use case, and the systems integration that makes a quantized payload a transparent bus primitive.

Resource and thermal control. OOM-driven batch-size search is folklore; batch-probe makes it robust (optimizer-state-aware peak-memory accounting, a unified multi-framework OOM-recovery path). Thermal management of CPUs and accelerators is usually fixed-threshold throttling; batch-probe instead estimates thermal state with a Kalman filter [6] and acts on the *predicted* trajectory, throttling parallelism before overshoot—closer to model-predictive control than to a thermostat.

X. ARTIFACT AVAILABILITY

All three projects are open and on PyPI—*nats-bursting* [12], *turboquant-pro* [13], and *batch-probe* [14]; Table V reports their shipped size. The benchmark harness reproduces every measurement here: *bench_compression.py* (Tier 1, with a real-embedding variant), *bench_transport.py* and the co-existing real-hop driver (Tier 2), and *bench_cluster.py* with `-mode cold|warm|scale` (Tier 3), the last gated behind `-i-have-checked-nrp-policy`. Result files and a *make_figures.py* that regenerates every figure are committed alongside.

XI. CONCLUSION

nats-bursting, *turboquant-pro*, and *batch-probe* form a small, composable, open pipeline that makes a shared, containerized cluster usable from inside the event loop—effectively, and within policy by construction. We described each system’s architecture, including the politeness controller, the compute-on-codes codec, and the Kalman-filtered thermal governor, and quantified the composition with real measurements, threats stated and remaining runs scoped. More broadly, the pattern—cluster pods as first-class subscribers reached over an outbound leaf link, behind a controller that defers to its neighbors—generalizes to any Kubernetes cluster a researcher can bridge to, and the policy-safe discipline it enforces is the right default for sharing a multi-tenant resource.

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